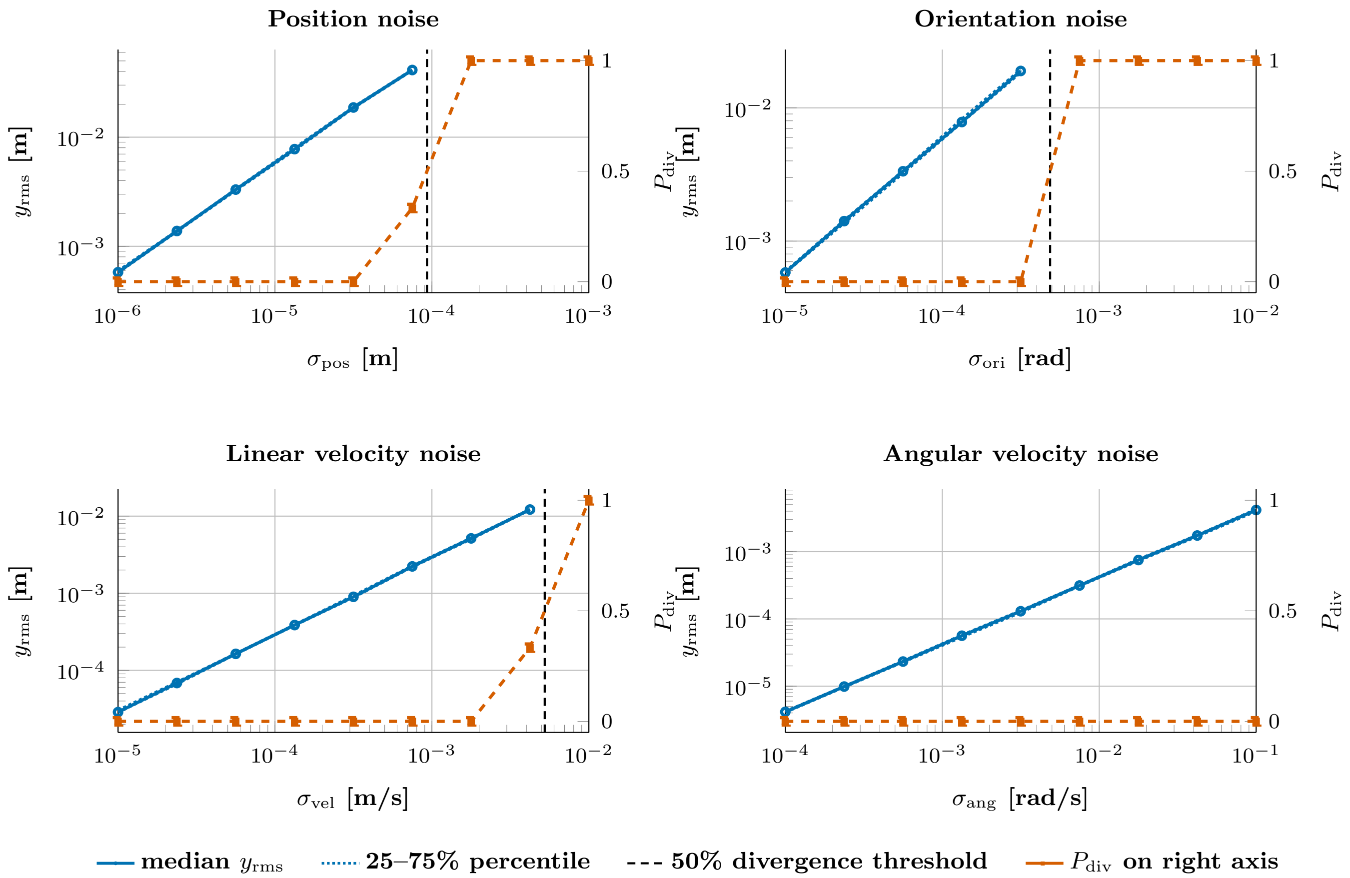
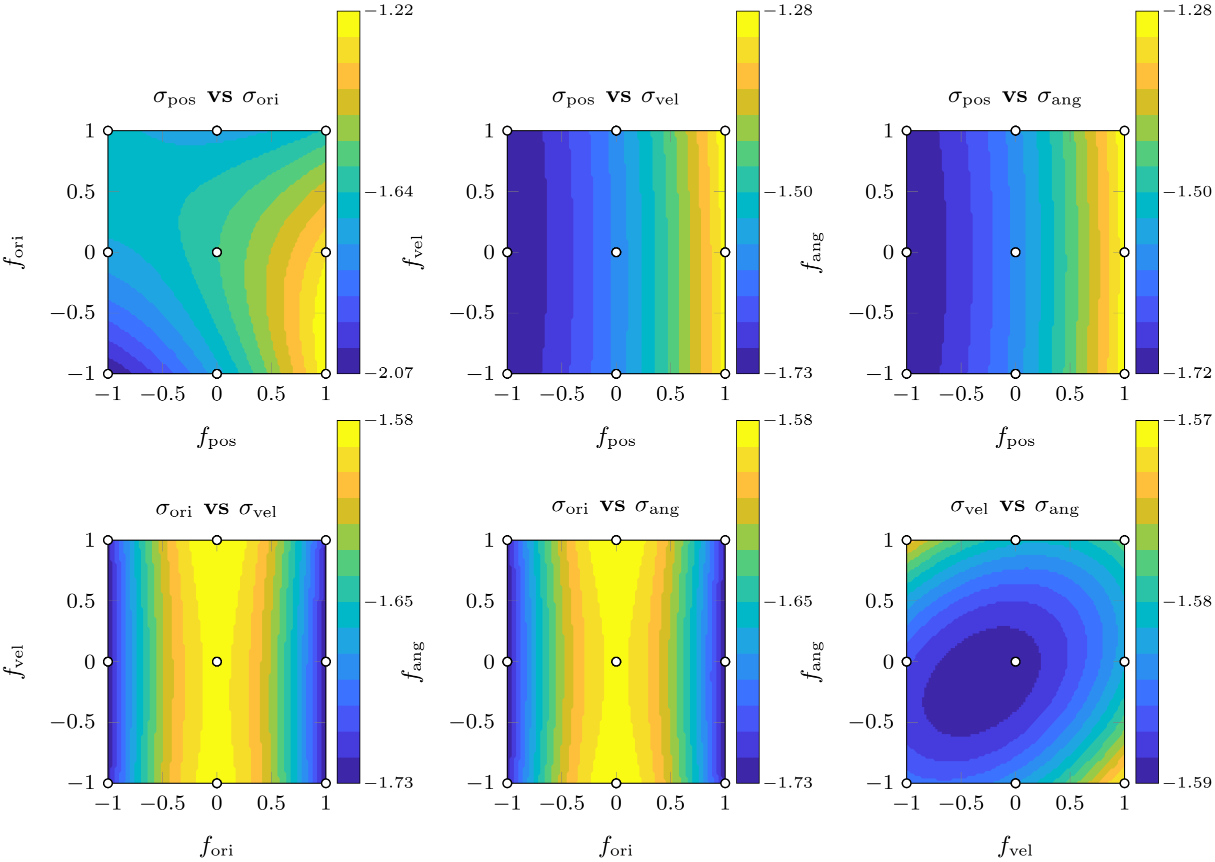
Per-factor noise sensitivity

y_{rms}: weighted RMS of state tracking error over the last 50% of each run; replicates per noise level: 3.



Response surface contours: $\log_{10}(y_{\text{rms}})$

Quadratic RSM fit; each panel scans two factors over $[-1,+1]$ with the other two held at coded 0.



f_i is the coded noise level (Box–Behnken design); $\sigma_i = 10^{\,\text{logc}_i + 0.5\,f_i}$ with $\text{logc} = (-4.5, -3.5, -3.5, -2.5)$, so each coded unit scales σ by $\sqrt{10} \approx 3.16\times$. $\circ$ Box–Behnken design points (projected onto each pair).